Bachelor Project

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Technical
University
in Prague

F3

Faculty of Electrical Engineering
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Design of a platform for 360° LiDAR

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Declaration

I hereby declare that artificial intelligence was used during the writing of this thesis, specifically as an aid in generating TikZ figures. Grammar was controlled with Grammarly. All generated content was verified, and I take full responsibility for the contents of this project. All information taken from the work of other authors has been properly cited.

In Prague, 13. November 2025

Abstract

A wide range of Light Detection and Ranging (LiDAR) sensors for spatial scanning is available on the market today; however, most remain too expensive for amateur use. The aim of this miniproject is to design a cost-effective platform for a 360° LiDAR sensor, which will be utilised in a subsequent bachelor's thesis to obtain a 3D model of the scanned environment. This work provides an overview of existing DIY solutions, evaluates their limitations and proposes a platform design that addresses the identified weaknesses. The result is a configurable platform that can be quickly reassembled depending on whether the user prioritises more complete coverage or quicker scan.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

Supervisor: Ing. Pavel Puričer, Ph.D.

Abstrakt

Dnes je na trhu široká nabídka senzorů pro prostorové skenování; většina z nich je však příliš drahá pro amatérské využití. Cílem tohoto miniprojektu je navrhnout cenově dostupnou platformu pro 360° LiDAR senzor, která bude v navazující bakalářské práci využita k vytvoření 3D modelu naskenovaného prostředí. Práce analyzuje již existující DIY řešení, identifikuje jejich omezení a navrhuje konstrukci platformy, která tyto nedostatky omezuje. Výsledkem je modularní platforma, kterou lze rychle upravit podle požadavků uživatele.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 360° LiDAR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localisation and Mapping

Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [1; 2, Ch. 4-6]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the cheapest commercial 3D LiDAR sensor, which was found during the research for this miniproject, costs $249 [3].

The primary motivation behind this mini-project (and the subsequent bachelor's thesis) is to develop a cost-effective platform for a 360° STL-19P LiDAR sensor that can be utilised for 3D spatial scanning and visualisation of the surrounding environment.

Figure 1.1: Photo of STL-19P LiDAR [1]

[1]Photo and parameters were taken from `https://www.waveshare.com/wiki/D500_LiDAR_Kit`

Table 1.1: Main parameters of STL-19P LiDAR[1]

Typical measuring range [m]	0,03–12
Angular resolution [°]	0,72
Dimensions(WxLxH) [mm, mm, mm]	38,59x38,59x33,50
Communication interface	UART @ 230400 bps

A subsequent bachelor's thesis will then focus on the Digital Signal Processing (DSP) techniques for the effective processing of the acquired point cloud data and proper utilisation of open-source visualisation tools (such as Python's Open3D library [4]) to recreate a 3D scene.

Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms discovered during the research for this project, analysing their pros and cons. Subsequently, the proposed design is presented, along with the chosen design decisions and their justifications.

2.1 Existing DIY Solutions for Environmental Mapping

2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilise a single-point LiDAR module mounted on a platform which can rotate and tilt. The implementation could be found in this project: [5].

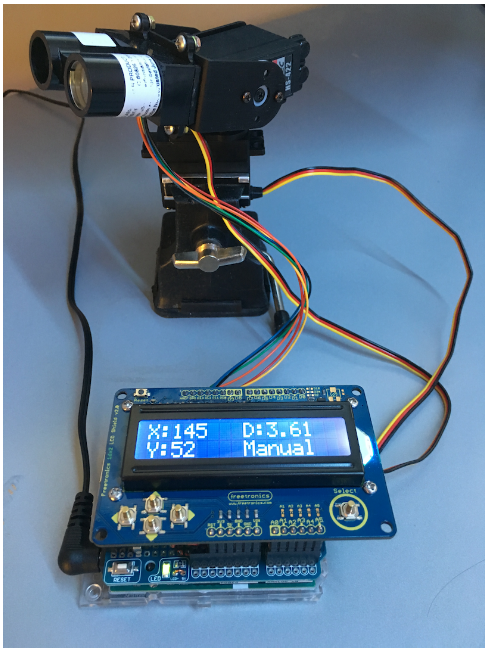


Figure 2.1: Implementation in project [5] using a single-point LiDAR

Pros

+ Single-point LiDARs are the most cost-effective among all existing types
+ Simple design

Cons

– Slow data acquisition relative to methods leveraging 2D LiDAR's speed
– Movement of lidar may introduce inacuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project: [6]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360-degree sweep. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

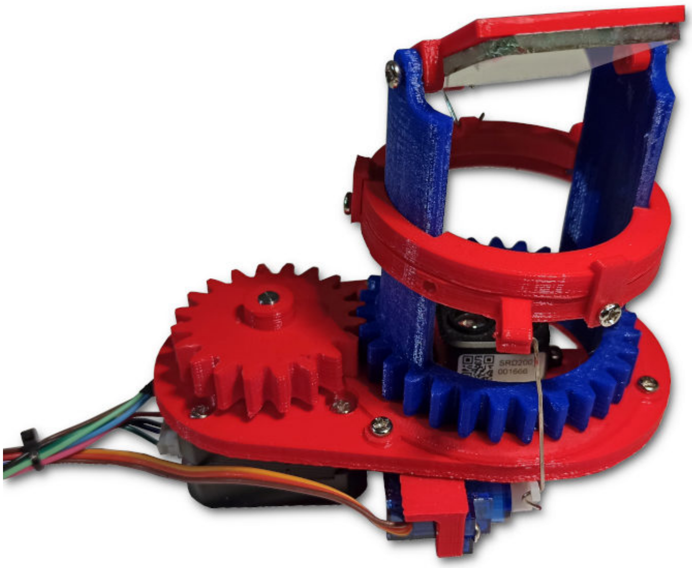


Figure 2.2: Implementation in project [6] using a single-point LiDAR and a mirror

Pros

- + Cost-effectiveness as in previous implementation
- + Simple design
- + Much easier to correct the distance measurement since the LiDAR is stationary and the distance from it to the mirror is known.

Cons

- – Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- – Mirror's quality may affect the accuracy of distance measurements
- – Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilises a 2D LiDAR sensor mounted on a platform that can tilt the sensor side to side, as demonstrated in this project [7]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning.

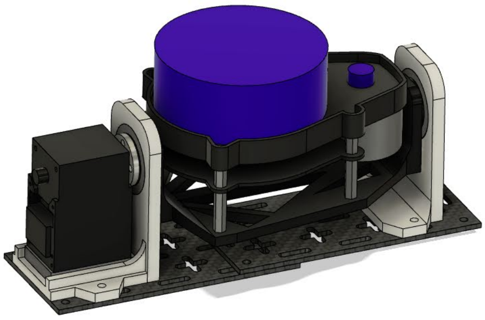


Figure 2.3: Implementation in project [7] using a tilting platform

Pros

+ Faster data acquisition compared to solutions with single-point LiDAR measurements
+ Higher resolution given by the use of a 2D LiDAR sensor
+ This design could be potentially used for Simultaneous Localisation and Mapping (SLAM) applications, given the higher speed

Cons

– More expensive than single-point LiDAR solutions
– Point distribution is non-uniform due to the tilting mechanism, which produces denser point clouds on the axis of tilting (see fig. 2.4)
– Mechanical design is more prone to vibrations produced by the LiDAR itself, affecting measurement accuracy

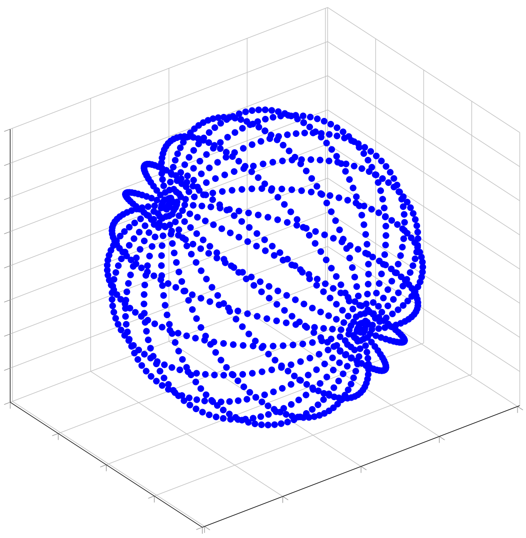


Figure 2.4: Simulation of an obtained point cloud using a tilting platform

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate around its axis.

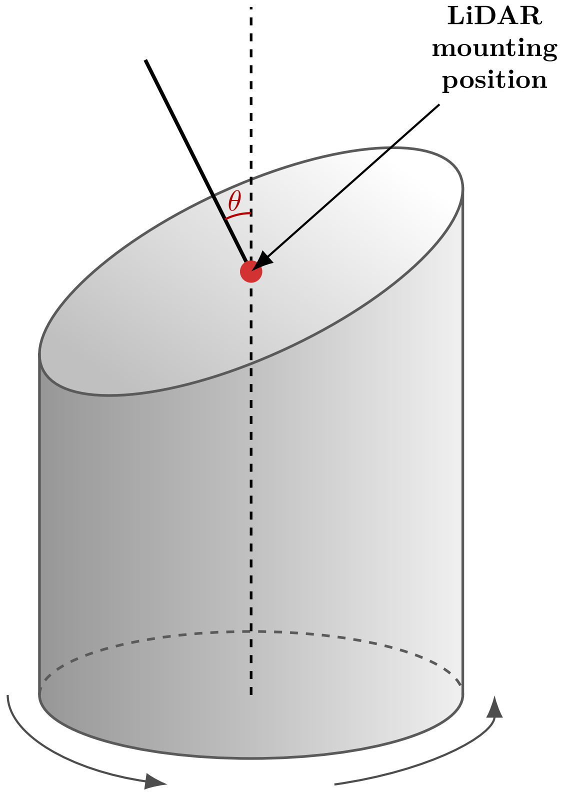
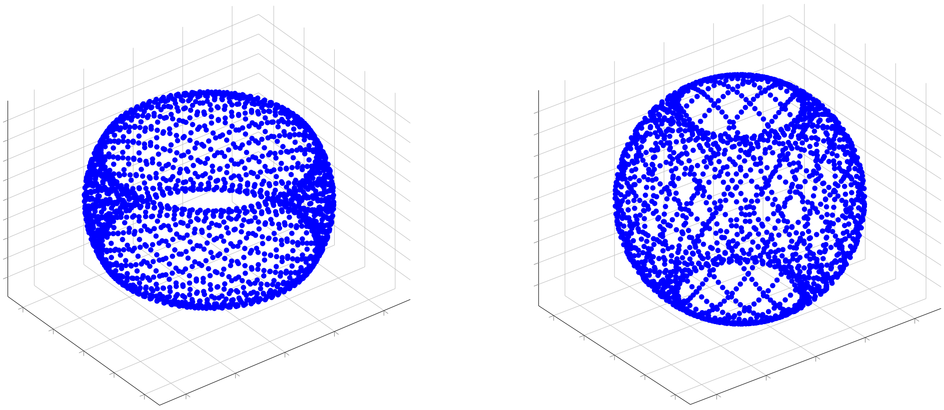


Figure 2.5: Proposed platform design. θ is the tilt angle

This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high resolution, while also eliminating some of the drawbacks of previous designs, namely:

+ The platform will be pretty rigid, minimising vibrations
+ Mechanical design is very simple, making it easy to build and troubleshoot
+ The resulting point cloud will be more uniformly distributed compared to the one obtained using the tilting platform design presented in section 2.1.3 unless the tilt angle is extremely steep (see fig. 2.6)

Such a design does have its drawbacks, however. The first one is immediately apparent from fig. 2.6: there are blind spots in the point cloud where no



(a) : $\theta = 30^{\circ}$ **(b) :** $\theta = 60^{\circ}$

Figure 2.6: Simulation of an obtained point cloud using a rotating tilted platform with two different tilt angles θ

points are measured. While these blind spots can be minimised by choosing a large tilt angle, this approach reintroduces the problem of non-uniform point cloud density.

Besides that, compared to the tilting platform design presented in section 2.1.3, the proposed design would need to rotate across twice as large a range to obtain full coverage of the environment, since the tilting platform scanned only within $[-90^{\circ}, +90^{\circ}]$, whereas the current design must complete a full 360-degree rotation. This would lead to longer scanning times.

2.2.1 Controlling the Platform

Since we want our platform to be precisely controlled, we need to choose an appropriate motor. Stepper motors are commonly used in applications requiring precise positioning, as they can be controlled to move in small, discrete steps. For this design, a NEMA 17 17HS8401 stepper motor is used to rotate the LiDAR platform.

Figure 2.7: NEMA 17 17HS8401 photo[2]

Table 2.1: Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor[2]

Motor Weight [Ω]	360
Holding Torque [N m]	0,5
Step Angle [°]	1,8
# of Steps per Revolution [-]	200
Rated Current [A]	1,68

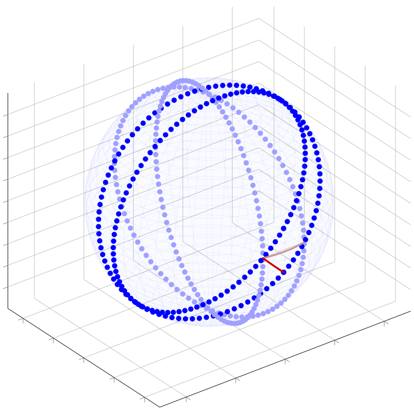
For heavy load applications, it is important to consider the potential error sources of stepper motors, such as the stepper motor losing steps due to a heavy load or by driving the motor faster than its maximum speed [8, p. 13]. However, in our case, the load will be relatively light for this motor, and the speed will be kept low; thus, a feedback mechanism for correcting lost steps is not required.

2.2.2 Choosing the Tilt Angle

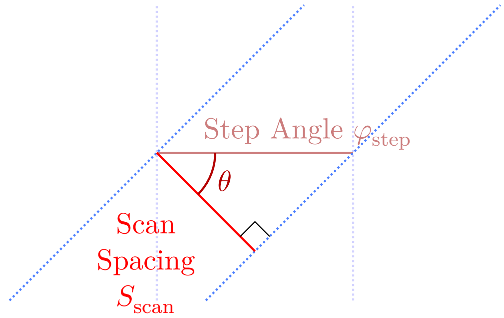
The tilt angle of the LiDAR platform is a crucial parameter that affects the size of the aforementioned blind spots in the point cloud (see fig. 2.6). However, increasing the tilt angle also increases the spacing between the scans when the platform is rotated in discrete steps (see fig. 2.8). It is obvious from the fig. 2.8b, that

$$S_{\text{scan}} = \varphi_{\text{step}} \cdot \sin(\theta), \tag{2.1}$$

[2]Photo and parameters were taken from `https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/`



(a) : 3D simulation



(b) : Simplified image showing the dependence of scan spacing S_{scan} on tilt angle θ

Figure 2.8: Visualisation of scan spacing dependence on the tilt angle

where the step angle φ_{step} is constant and is equal to 1.8° for our motor (see table 2.1). The following graph shows how the spacing changes with the tilt angle, providing our LiDAR's resolution as a reference value, which we would like to achieve by choosing a correct tilt angle:

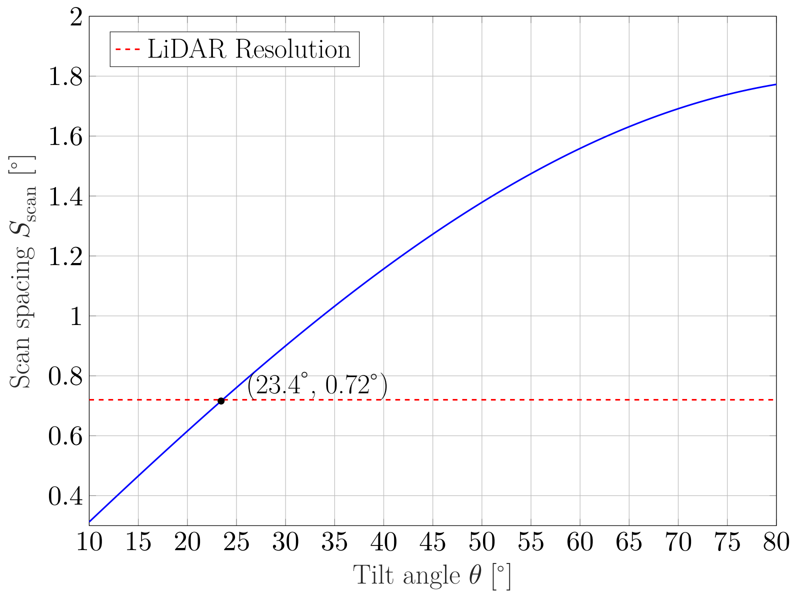


Figure 2.9: Scan spacing depending on the tilt angle θ

As could be seen from fig. 2.9, the spacing will be the same as the angular resolution of the LiDAR itself at the tilt angle of approximately 23.4°.

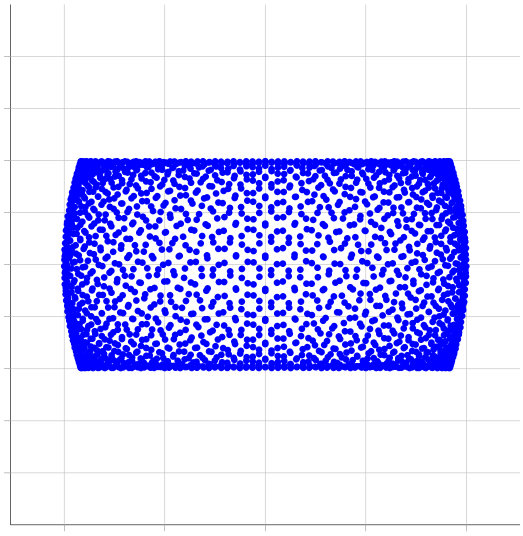


Figure 2.10: Side profile of a point cloud obtained using a rotating tilted platform with tilt angle $\theta = 23.4^{\circ}$

Examining the side profile of the point cloud obtained using this tilt angle, we can see that the blind spots are extremely large, making this tilt angle impractical for our application. Thankfully, there is a way to achieve a more complete coverage of the environment without sacrificing point cloud density.

2.2.3 Choosing a Gear Ratio for the Rotating Platform

To eliminate the need for a compromise between point cloud coverage and density when using a stepper motor to rotate the platform, we can incorporate a gear mechanism into our design. Using gears with a specific gear ratio allows us to choose a tilt angle that minimises blind spots while still producing a dense point cloud. The scan spacing will then be given by the following equation:

$$S_{\text{scan}} = R_{\text{gear}} \cdot \varphi_{\text{step}} \cdot \sin(\theta), \tag{2.2}$$

where R_{gear} is the ratio of the number of teeth on the driven gear to the number of teeth on the driving gear. A following graph is similar to fig. 2.9, but the third dimension represents different gear ratios:

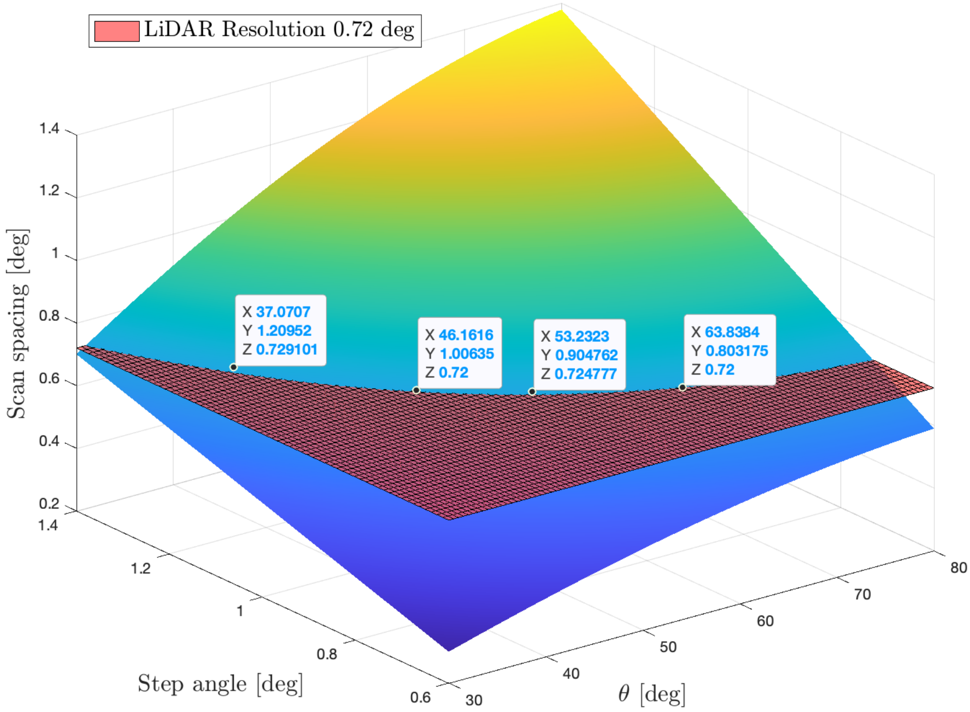


Figure 2.11: Point cloud resolution depending on the tilt angle θ and gear ratio

As can be seen from fig. 2.11, using a gearbox allows us to choose from a wide range of tilt angles without compromising on the spacing between scans.

Table 2.2: Best combinations for step angle/gear ratio/tilt angle

Step Angle [°]	Gear Ratio [-]	Tilt Angle [°]
1,2	3:2	37
1,0	9:5	46
0,9	2:1	53
0,8	9:4	64

Of course, by using the gearbox, we increase the number of steps required for a complete revolution, thereby increasing the scanning time. Nevertheless, by providing platforms of different sizes with appropriate gear ratios, we can create a modular platform that can be reassembled depending on the application.

Chapter 3

Conclusion

In summary, the proposed rotating tilted platform design addresses most of the drawbacks of the current DIY solutions presented at section 2.1: scanning time is relatively fast thanks to LiDAR's type, the platform is rigid, simple in its design, and produces a more uniform point cloud, granted that a suitable tilt angle is used.

The introduction of a gearbox represents a key innovation, as it provides the user with control over the trade-off between speed, point cloud density, and environmental coverage. Thus, the resulting platform achieves the best overall compromise among DIY solutions while remaining affordable, as all the parts can be 3D-printed due to their simple design. This makes such a platform ideal for an amateur use case, which is what this project meant to achieve.

Appendix A

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